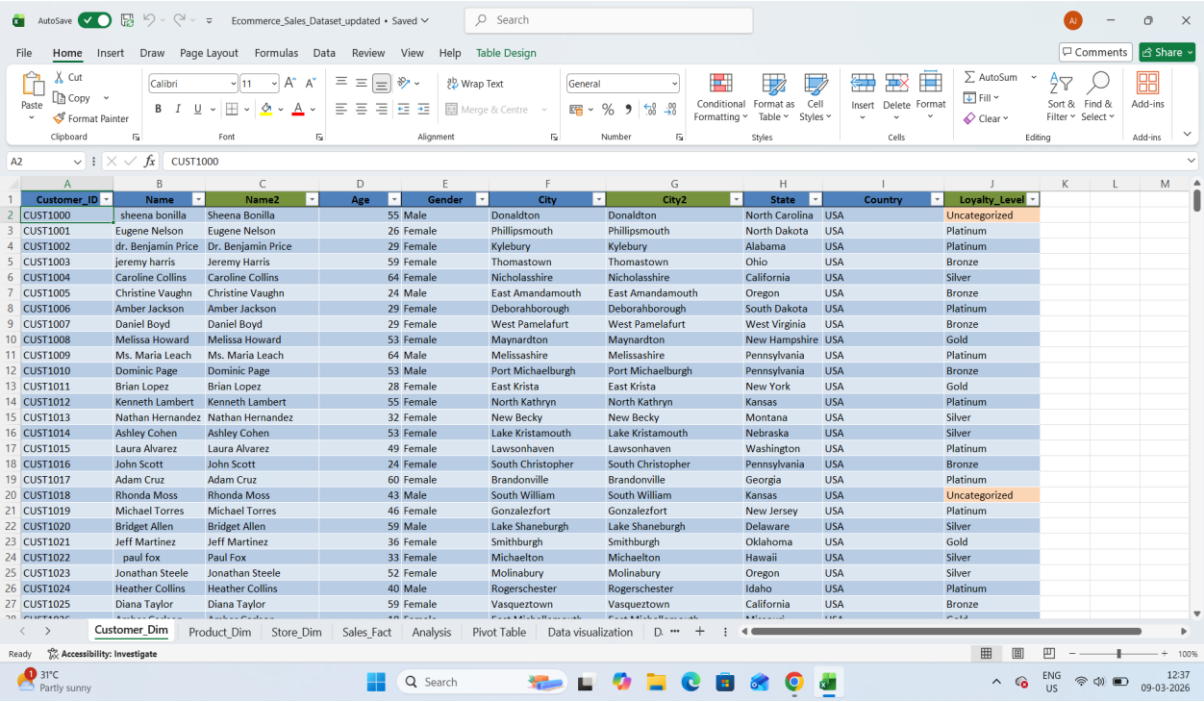
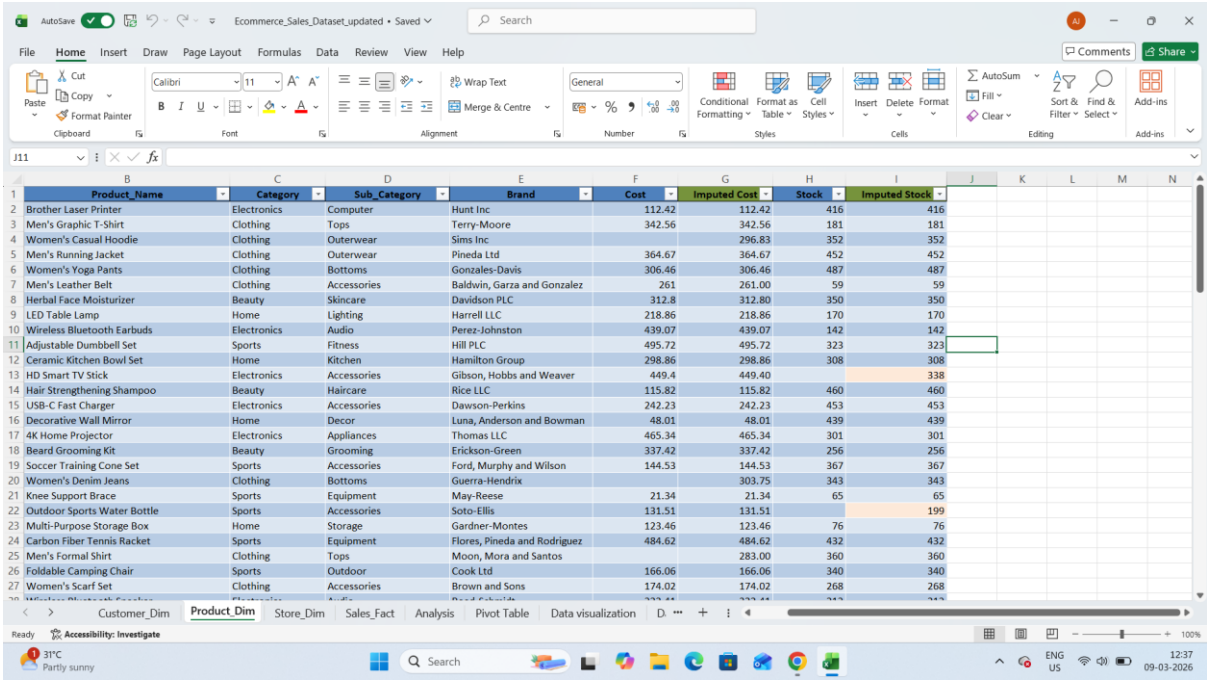
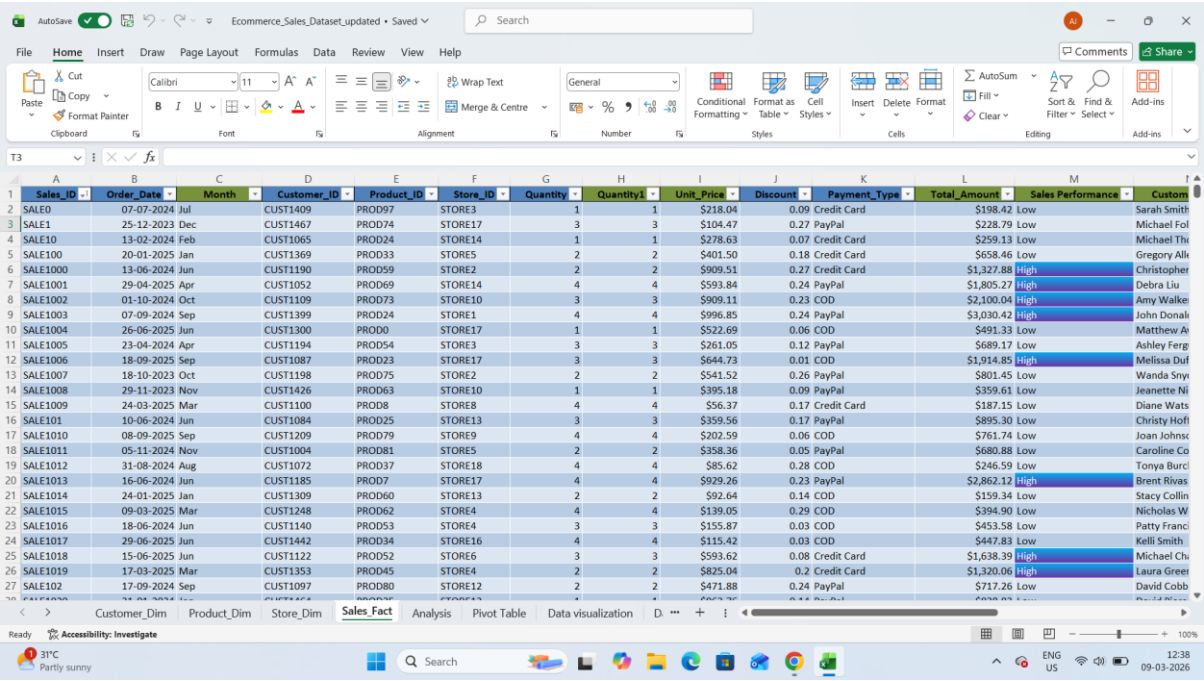
MODULE 1 END ASSIGNMENT



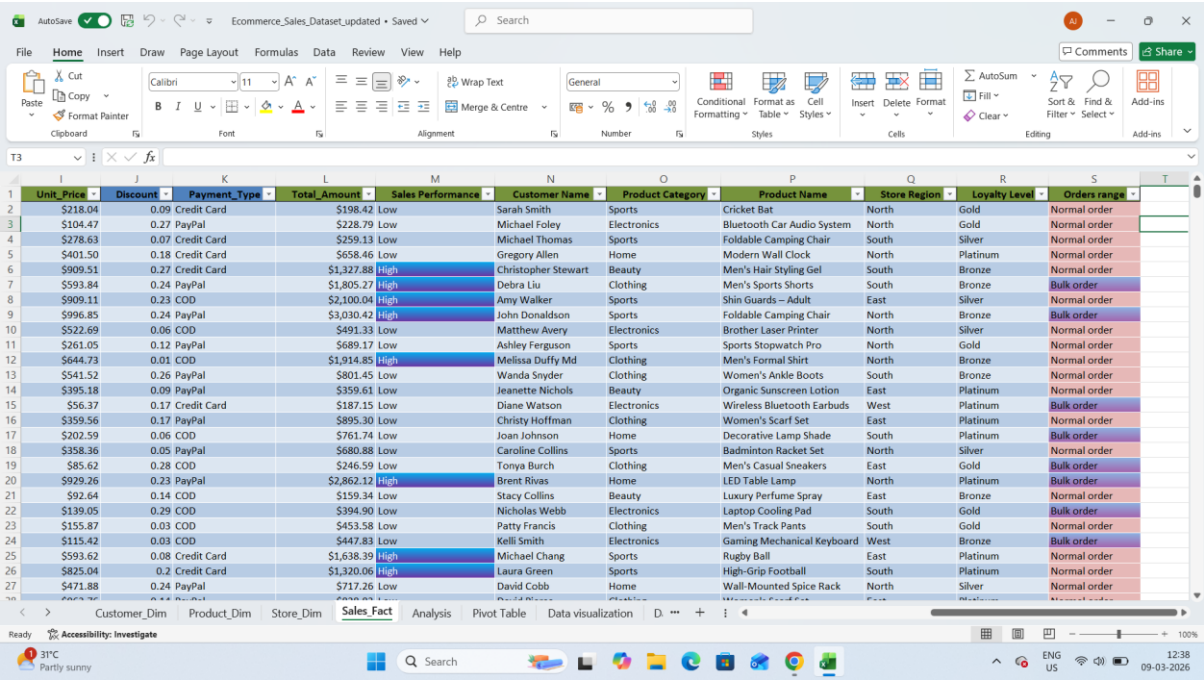
Customer_ID	Name	Name2	Age	Gender	City	City2	State	Country	Loyalty_Level
CUST1000	sheena bonilla	Sheena Bonilla	55	Male	Donaldton	Donaldton	North Carolina	USA	Uncategorized
CUST1001	Eugene Nelson	Eugene Nelson	26	Female	Phillipsmouth	Phillipsmouth	North Dakota	USA	Platinum
CUST1002	dr. Benjamin Price	Dr. Benjamin Price	29	Female	Kylebury	Kylebury	Alabama	USA	Platinum
CUST1003	jeremy harris	Jeremy Harris	59	Female	Thomastown	Thomastown	Ohio	USA	Bronze
CUST1004	Caroline Collins	Caroline Collins	64	Female	Nicholasshire	Nicholasshire	California	USA	Silver
CUST1005	Christine Vaughn	Christine Vaughn	24	Male	East Amandamouth	East Amandamouth	Oregon	USA	Bronze
CUST1006	Amber Jackson	Amber Jackson	29	Female	Deborahborough	Deborahborough	South Dakota	USA	Platinum
CUST1007	Daniel Boyd	Daniel Boyd	29	Female	West Pamelafurt	West Pamelafurt	West Virginia	USA	Bronze
CUST1008	Melissa Howard	Melissa Howard	53	Female	Maynardton	Maynardton	New Hampshire	USA	Gold
CUST1009	Ms. Maria Leach	Ms. Maria Leach	64	Male	Melissashire	Melissashire	Pennsylvania	USA	Platinum
CUST1010	Dominic Page	Dominic Page	53	Male	Port Michaelburgh	Port Michaelburgh	Pennsylvania	USA	Bronze
CUST1011	Brian Lopez	Brian Lopez	28	Female	East Krista	East Krista	New York	USA	Gold
CUST1012	Kenneth Lambert	Kenneth Lambert	55	Female	North Kathryn	North Kathryn	Kansas	USA	Platinum
CUST1013	Nathan Hernandez	Nathan Hernandez	32	Female	New Becky	New Becky	Montana	USA	Silver
CUST1014	Ashley Cohen	Ashley Cohen	53	Female	Lake Kristamouth	Lake Kristamouth	Nebraska	USA	Silver
CUST1015	Laura Alvarez	Laura Alvarez	49	Female	Lawsonhaven	Lawsonhaven	Washington	USA	Platinum
CUST1016	John Scott	John Scott	24	Female	South Christopher	South Christopher	Pennsylvania	USA	Bronze
CUST1017	Adam Cruz	Adam Cruz	60	Female	Brandonville	Brandonville	Georgia	USA	Platinum
CUST1018	Rhonda Moss	Rhonda Moss	43	Male	South William	South William	Kansas	USA	Uncategorized
CUST1019	Michael Torres	Michael Torres	46	Female	Gonzalezfort	Gonzalezfort	New Jersey	USA	Platinum
CUST1020	Bridget Allen	Bridget Allen	59	Male	Lake Shaneburgh	Lake Shaneburgh	Delaware	USA	Silver
CUST1021	Jeff Martinez	Jeff Martinez	36	Female	Smithburgh	Smithburgh	Oklahoma	USA	Gold
CUST1022	paul fox	Paul Fox	33	Female	Michaelton	Michaelton	Hawaii	USA	Silver
CUST1023	Jonathan Steele	Jonathan Steele	52	Female	Molinabury	Molinabury	Oregon	USA	Silver
CUST1024	Heather Collins	Heather Collins	40	Male	Rogerschester	Rogerschester	Idaho	USA	Platinum
CUST1025	Diana Taylor	Diana Taylor	59	Female	Vasqueztown	Vasqueztown	California	USA	Bronze



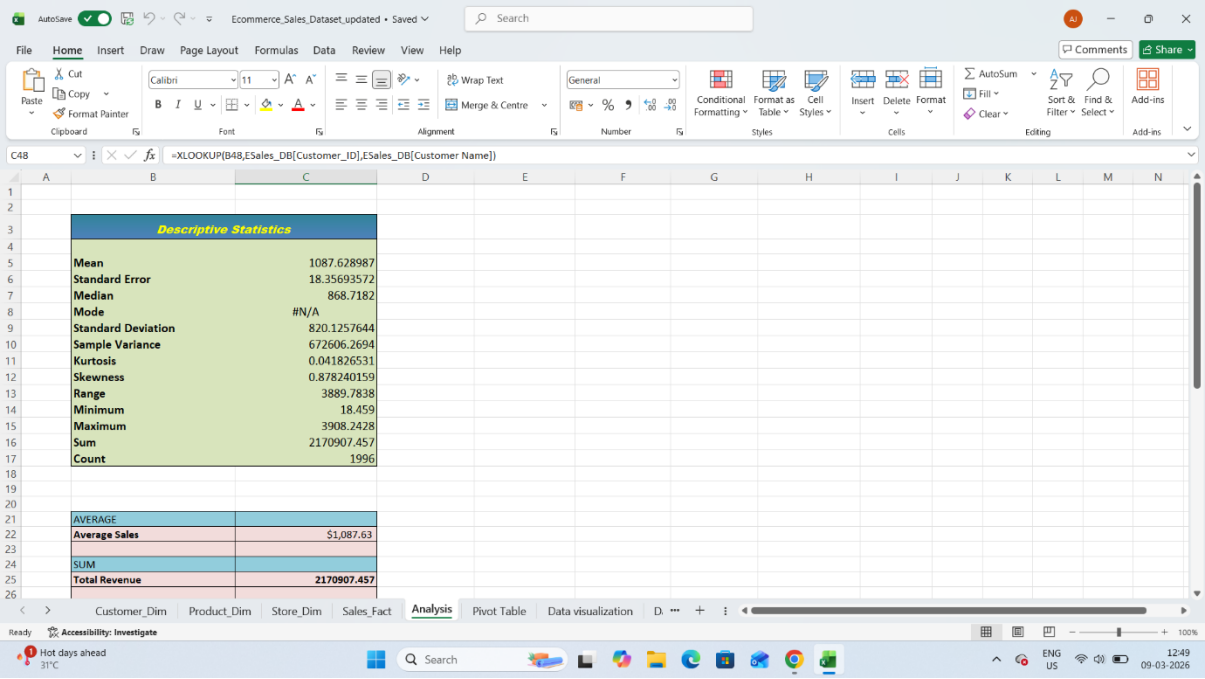
Product_Name	Category	Sub_Category	Brand	Cost	Imputed Cost	Stock	Imputed Stock
Brother Laser Printer	Electronics	Computer	Hunt Inc	112.42	112.42	416	416
Men's Graphic T-Shirt	Clothing	Tops	Terry-Moore	342.56	342.56	181	181
Women's Casual Hoodie	Clothing	Outerwear	Sims Inc		296.83	352	352
Men's Running Jacket	Clothing	Outerwear	Pineda Ltd	364.67	364.67	452	452
Women's Yoga Pants	Clothing	Bottoms	Gonzales-Davis	306.46	306.46	487	487
Men's Leather Belt	Clothing	Accessories	Baldwin, Garza and Gonzalez	261	261.00	59	59
Herbal Face Moisturizer	Beauty	Skincare	Davidson PLC	312.8	312.80	350	350
LED Table Lamp	Home	Lighting	Harrell LLC	218.86	218.86	170	170
Wireless Bluetooth Earbuds	Electronics	Audio	Perez-Johnston	439.07	439.07	142	142
Adjustable Dumbbell Set	Sports	Fitness	Hill PLC	495.72	495.72	323	323
Ceramic Kitchen Bowl Set	Home	Kitchen	Hamilton Group	298.86	298.86	308	308
HD Smart TV Stick	Electronics	Accessories	Gibson, Hobbs and Weaver	449.4	449.40		338
Hair Strengthening Shampoo	Beauty	Haircare	Rice LLC	115.82	115.82	460	460
USB-C Fast Charger	Electronics	Accessories	Dawson-Perkins	242.23	242.23	453	453
Decorative Wall Mirror	Home	Decor	Luna, Anderson and Bowman	48.01	48.01	439	439
4K Home Projector	Electronics	Appliances	Thomas LLC	465.34	465.34	301	301
Beard Grooming Kit	Beauty	Grooming	Erickson-Green	337.42	337.42	256	256
Soccer Training Cone Set	Sports	Accessories	Ford, Murphy and Wilson	144.53	144.53	367	367
Women's Denim Jeans	Clothing	Bottoms	Guerra-Hendrix		303.75	343	343
Knee Support Brace	Sports	Equipment	May-Reese	21.34	21.34	65	65
Outdoor Sports Water Bottle	Sports	Accessories	Soto-Ellis	131.51	131.51		199
Multi-Purpose Storage Box	Home	Storage	Gardner-Montes	123.46	123.46	76	76
Carbon Fiber Tennis Racket	Sports	Equipment	Flores, Pineda and Rodriguez	484.62	484.62	432	432
Men's Formal Shirt	Clothing	Tops	Moon, Mora and Santos		283.00	360	360
Foldable Camping Chair	Sports	Outdoor	Cook Ltd	166.06	166.06	340	340
Women's Scarf Set	Clothing	Accessories	Brown and Sons	174.02	174.02	268	268



Sales_ID	Order_Date	Month	Customer_ID	Product_ID	Store_ID	Quantity	Quantity1	Unit_Price	Discount	Payment_Type	Total_Amount	Sales Performance	Custom
SALE0	07-07-2024	Jul	CUST1409	PROD97	STORE3	1	1	$218.04	0.09	Credit Card	$198.42	Low	Sarah Smith
SALE1	25-12-2023	Dec	CUST1467	PROD74	STORE17	3	3	$104.47	0.27	PayPal	$228.79	Low	Michael Fol
SALE10	13-02-2024	Feb	CUST1065	PROD24	STORE14	1	1	$278.63	0.07	Credit Card	$259.13	Low	Michael Th
SALE100	20-01-2025	Jan	CUST1369	PROD33	STORE5	2	2	$401.50	0.18	Credit Card	$658.46	Low	Gregory All
SALE1000	13-06-2024	Jun	CUST1190	PROD59	STORE2	2	2	$909.51	0.27	Credit Card	$1,327.88	High	Christopher
SALE1001	29-04-2025	Apr	CUST1052	PROD69	STORE14	4	4	$593.84	0.24	PayPal	$1,805.27	High	Debra Liu
SALE1002	01-10-2024	Oct	CUST1109	PROD73	STORE10	3	3	$909.11	0.23	COD	$2,100.04	High	Amy Walke
SALE1003	07-09-2024	Sep	CUST1399	PROD24	STORE1	4	4	$996.85	0.24	PayPal	$3,030.42	High	John Donal
SALE1004	26-06-2025	Jun	CUST1300	PROD0	STORE17	1	1	$522.69	0.06	COD	$491.33	Low	Matthew A
SALE1005	23-04-2024	Apr	CUST1194	PROD54	STORE3	3	3	$261.05	0.12	PayPal	$689.17	Low	Ashley Ferg
SALE1006	18-09-2025	Sep	CUST1087	PROD23	STORE17	3	3	$644.73	0.01	COD	$1,914.85	High	Melissa Duf
SALE1007	18-10-2023	Oct	CUST1198	PROD75	STORE2	2	2	$541.52	0.26	PayPal	$801.45	Low	Wanda Sny
SALE1008	29-11-2023	Nov	CUST1426	PROD63	STORE10	1	1	$395.18	0.09	PayPal	$359.61	Low	Jeanette Ni
SALE1009	24-03-2025	Mar	CUST1100	PROD8	STORE8	4	4	$56.37	0.17	Credit Card	$187.15	Low	Diane Wats
SALE101	10-06-2024	Jun	CUST1084	PROD25	STORE13	3	3	$359.56	0.17	PayPal	$895.30	Low	Christy Hof
SALE1010	08-09-2025	Sep	CUST1209	PROD79	STORE9	4	4	$202.59	0.06	COD	$761.74	Low	Joan Johns
SALE1011	05-11-2024	Nov	CUST1004	PROD81	STORE5	2	2	$358.36	0.05	PayPal	$680.88	Low	Caroline Co
SALE1012	31-08-2024	Aug	CUST1072	PROD37	STORE18	4	4	$85.62	0.28	COD	$246.59	Low	Tonya Burc
SALE1013	16-06-2024	Jun	CUST1185	PROD7	STORE17	4	4	$929.26	0.23	PayPal	$2,862.12	High	Brent Rivas
SALE1014	24-01-2025	Jan	CUST1309	PROD60	STORE13	2	2	$92.64	0.14	COD	$159.34	Low	Stacy Collin
SALE1015	09-03-2025	Mar	CUST1248	PROD62	STORE4	4	4	$139.05	0.29	COD	$394.90	Low	Nicholas W
SALE1016	18-06-2024	Jun	CUST1140	PROD53	STORE4	3	3	$155.87	0.03	COD	$453.58	Low	Patty Franc
SALE1017	29-06-2025	Jun	CUST1442	PROD34	STORE16	4	4	$115.42	0.03	COD	$447.83	Low	Kelli Smith
SALE1018	15-06-2025	Jun	CUST1122	PROD52	STORE6	3	3	$593.62	0.08	Credit Card	$1,638.39	High	Michael Ch
SALE1019	17-03-2025	Mar	CUST1353	PROD45	STORE4	2	2	$825.04	0.2	Credit Card	$1,320.06	High	Laura Gree
SALE102	17-09-2024	Sep	CUST1097	PROD80	STORE12	2	2	$471.88	0.24	PayPal	$717.26	Low	David Cobb



Unit_Price	Discount	Payment_Type	Total_Amount	Sales Performance	Customer Name	Product Category	Product Name	Store Region	Loyalty Level	Orders range
$218.04	0.09	Credit Card	$198.42	Low	Sarah Smith	Sports	Cricket Bat	North	Gold	Normal order
$104.47	0.27	PayPal	$228.79	Low	Michael Foley	Electronics	Bluetooth Car Audio System	North	Gold	Normal order
$278.63	0.07	Credit Card	$259.13	Low	Michael Thomas	Sports	Foldable Camping Chair	South	Silver	Normal order
$401.50	0.18	Credit Card	$658.46	Low	Gregory Allen	Home	Modern Wall Clock	North	Platinum	Normal order
$909.51	0.27	Credit Card	$1,327.88	High	Christopher Stewart	Beauty	Men's Hair Styling Gel	South	Bronze	Normal order
$593.84	0.24	PayPal	$1,805.27	High	Debra Liu	Clothing	Men's Sports Shorts	South	Bronze	Bulk order
$909.11	0.23	COD	$2,100.04	High	Amy Walker	Sports	Shin Guards – Adult	East	Silver	Normal order
$996.85	0.24	PayPal	$3,030.42	High	John Donaldson	Sports	Foldable Camping Chair	North	Bronze	Bulk order
$522.69	0.06	COD	$491.33	Low	Matthew Avery	Electronics	Brother Laser Printer	North	Silver	Normal order
$261.05	0.12	PayPal	$689.17	Low	Ashley Ferguson	Sports	Sports Stopwatch Pro	North	Gold	Normal order
$644.73	0.01	COD	$1,914.85	High	Melissa Duffy Md	Clothing	Men's Formal Shirt	North	Bronze	Normal order
$541.52	0.26	PayPal	$801.45	Low	Wanda Snyder	Clothing	Women's Ankle Boots	South	Bronze	Normal order
$395.18	0.09	PayPal	$359.61	Low	Jeanette Nichols	Beauty	Organic Sunscreen Lotion	East	Platinum	Normal order
$56.37	0.17	Credit Card	$187.15	Low	Diane Watson	Electronics	Wireless Bluetooth Earbuds	West	Platinum	Bulk order
$359.56	0.17	PayPal	$895.30	Low	Christy Hoffman	Clothing	Women's Scarf Set	East	Platinum	Normal order
$202.59	0.06	COD	$761.74	Low	Joan Johnson	Home	Decorative Lamp Shade	South	Platinum	Bulk order
$358.36	0.05	PayPal	$680.88	Low	Caroline Collins	Sports	Badminton Racket Set	North	Silver	Normal order
$85.62	0.28	COD	$246.59	Low	Tonya Burch	Clothing	Men's Casual Sneakers	East	Gold	Bulk order
$929.26	0.23	PayPal	$2,862.12	High	Brent Rivas	Home	LED Table Lamp	North	Platinum	Bulk order
$92.64	0.14	COD	$159.34	Low	Stacy Collins	Beauty	Luxury Perfume Spray	East	Bronze	Normal order
$139.05	0.29	COD	$394.90	Low	Nicholas Webb	Electronics	Laptop Cooling Pad	South	Gold	Bulk order
$155.87	0.03	COD	$453.58	Low	Patty Francis	Clothing	Men's Track Pants	South	Gold	Normal order
$115.42	0.03	COD	$447.83	Low	Kelli Smith	Electronics	Gaming Mechanical Keyboard	West	Bronze	Bulk order
$593.62	0.08	Credit Card	$1,638.39	High	Michael Chang	Sports	Rugby Ball	East	Platinum	Normal order
$825.04	0.2	Credit Card	$1,320.06	High	Laura Green	Sports	High-Grip Football	South	Platinum	Normal order
$471.88	0.24	PayPal	$717.26	Low	David Cobb	Home	Wall-Mounted Spice Rack	North	Silver	Normal order



Descriptive Statistics	
Mean	1087.628987
Standard Error	18.35693572
Median	868.7182
Mode	#N/A
Standard Deviation	820.1257644
Sample Variance	672606.2694
Kurtosis	0.041826531
Skewness	0.878240159
Range	3889.7838
Minimum	18.459
Maximum	3908.2428
Sum	2170907.457
Count	1996

AVERAGE	
Average Sales	$1,087.63
SUM	
Total Revenue	**2170907.457**

AVERAGE	
Average Sales	$1,087.63
SUM	
Total Revenue	**2170907.457**
MAXIMUM	
Maximum sale	**3908.2428**
COUNTIF	
Number of Bulk orders	**481**
Number of High sales	**893**
COUNTA	
Total Orders	**2000**
TEXT	
Month derivation from order date	**January**

Dropdown list using Data validation and deriving details using Xlookup formula

Customer ID	Customer Name	Loyalty Level	Product category	Quantity	Total Amount	Sales Performance	Region
CUST1344	elyn Brown	Gold	Home	1	$756.77	Low	North

Index and Match

Amy Walker	PROD73

Category wise Total Revenue

Category	Sum of Total_Amount
Beauty	321362.090
Clothing	377055.374
Electronics	504311.348
Home	427295.804
Sports	540882.842
Grand Total	**2170907.457**

Region wise Sales performance

Region	Sum of Total_Amount
East	619851.205
North	645166.750
South	635213.096
West	270676.406
Grand Total	**2170907.457**

Loyalty Level Analysis

Loyalty Level	Sum of Total_Amount
Bronze	522623.3819
Gold	516196.2792
Platinum	654777.5946
Silver	459797.8664
Uncategorized	17512.3353
Grand Total	**2170907.457**

Payment Type

Payment Type	Count of Sales_ID
COD	660
Credit Card	645
PayPal	695
Grand Total	**2000**

Month wise Trends

Month	Sum of Total_Amount
Jan	174449.14
Feb	155073.94
Mar	210704.52
Apr	192864.04
May	146511.96
Jun	162640.02
Jul	194512.58
Aug	191032.40
Sep	187144.40
Oct	186168.21
Nov	187386.17
Dec	182420.09
Grand Total	**2170907.457**

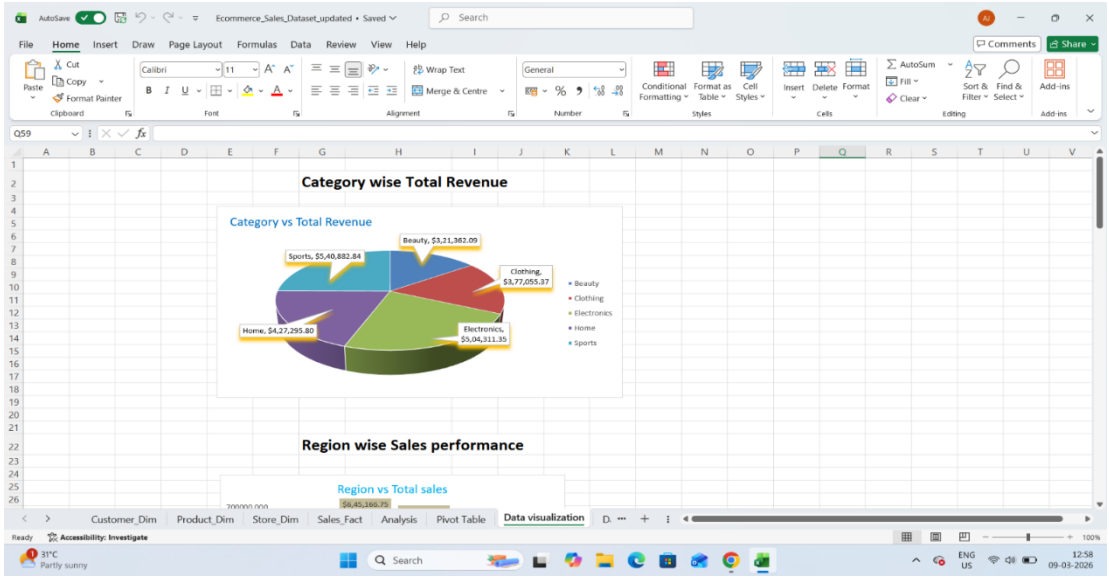
Top Products

Product name	Sum of Quantity1
Men's Casual Sneakers	87
Gaming Mechanical Keyboard	84
Portable Power Bank	84
Shin Guards – Adult	77

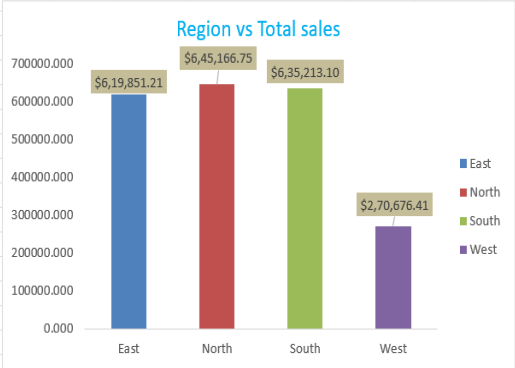
Customer_Dim | Product_Dim | Store_Dim | Sales_Fact | Analysis | Pivot Table

Top Products

Product name	Sum of Quantity1
Men's Casual Sneakers	87
Gaming Mechanical Keyboard	84
Portable Power Bank	84
Shin Guards – Adult	77
Wireless Mouse	73
4K Home Projector	72
Soccer Training Cone Set	70
Vitamin E Body Lotion	68
Face Foundation Kit	68
Laptop Cooling Pad	68
Grand Total	**751**

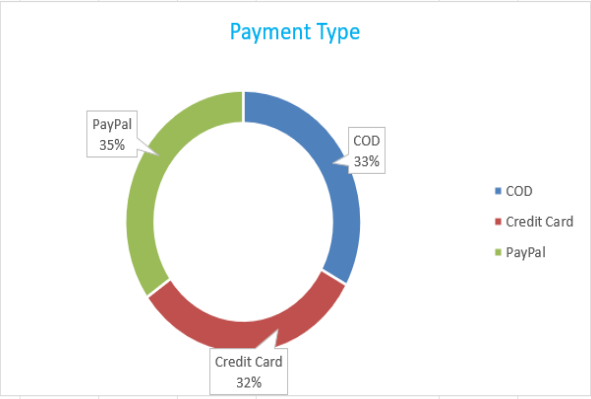


Region wise Sales performance

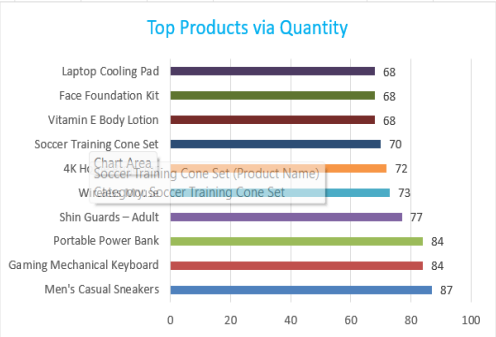


Loyalty Level Analysis

Payment Type



Top Products



Month wise Trends

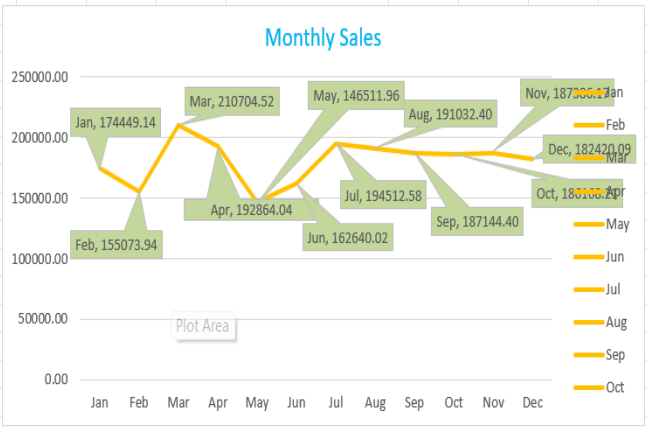
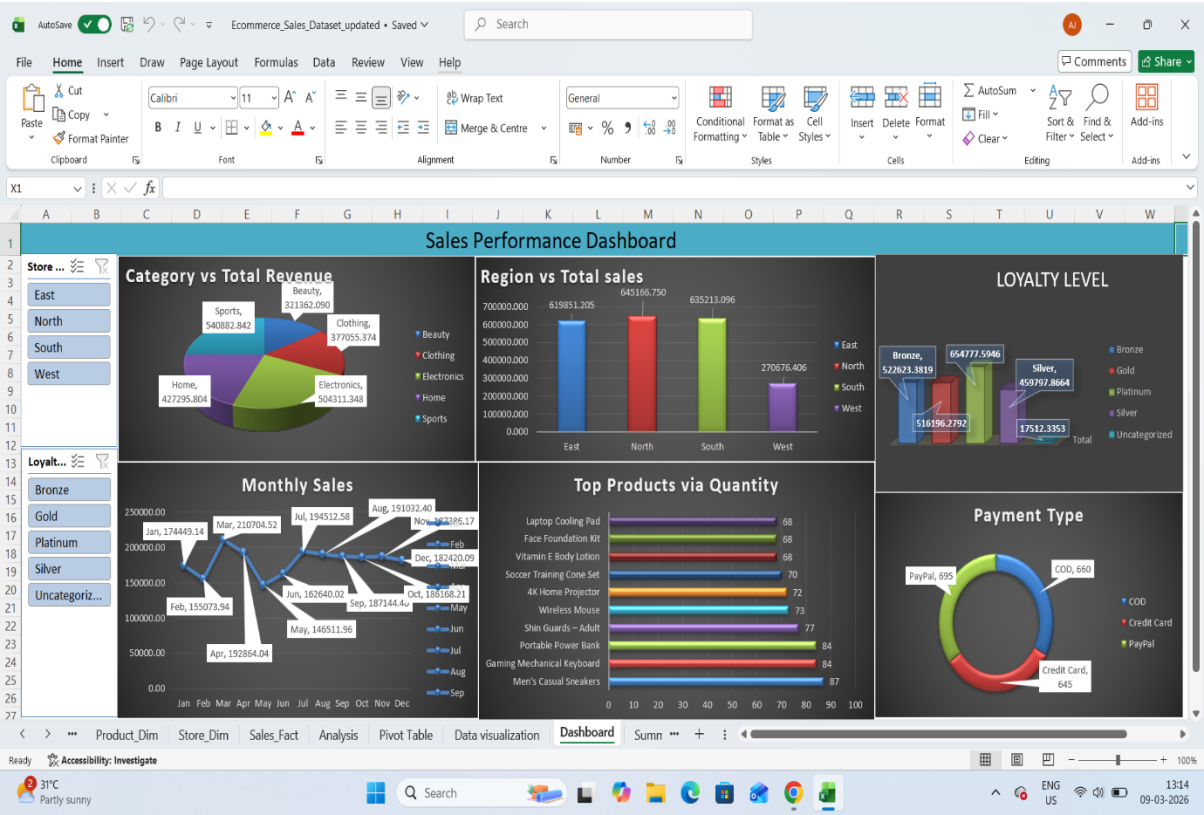
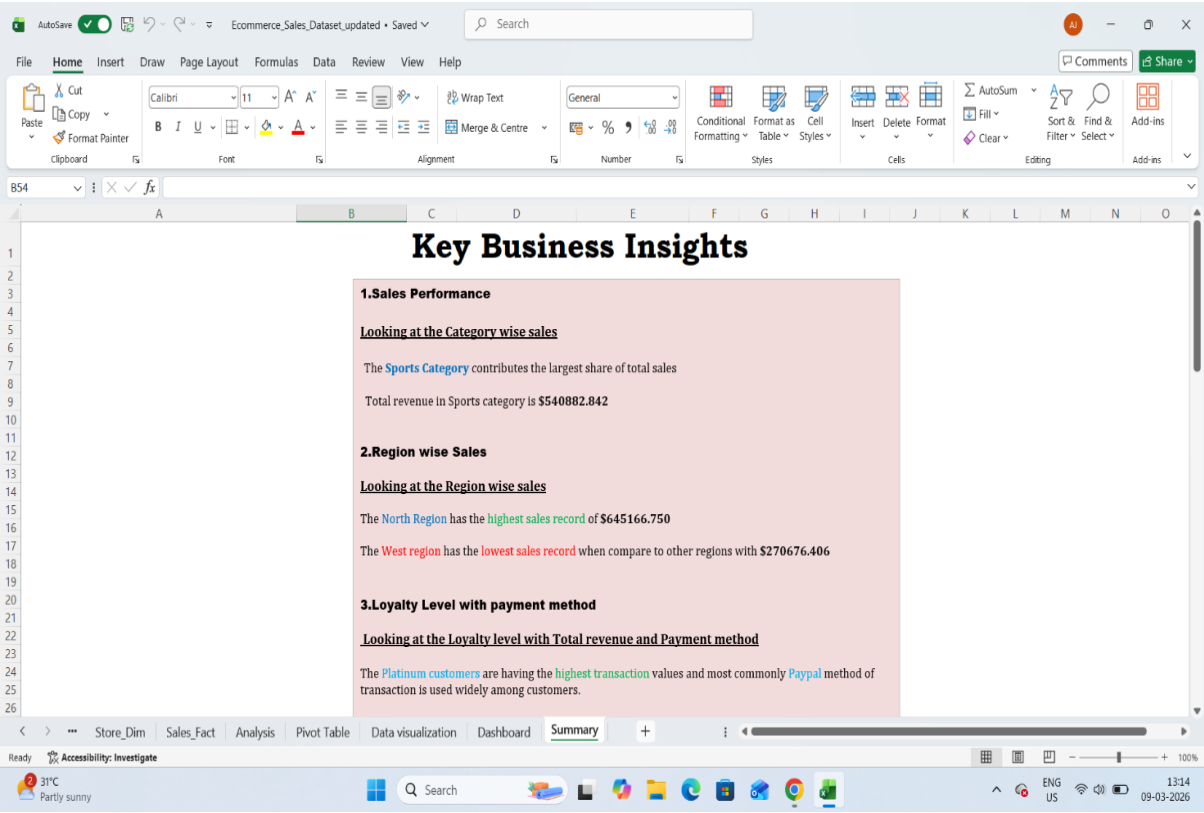


Chart Area





Key Business Insights

1.Sales Performance

Looking at the Category wise sales

The **Sports Category** contributes the largest share of total sales

Total revenue in Sports category is **$540882.842**

2.Region wise Sales

Looking at the Region wise sales

The North Region has the highest sales record of **$645166.750**

The West region has the lowest sales record when compare to other regions with **$270676.406**

3.Loyalty Level with payment method

Looking at the Loyalty level with Total revenue and Payment method

The Platinum customers are having the highest transaction values and most commonly Paypal method of transaction is used widely among customers.

Chart Area

3.Loyalty Level with payment method

Looking at the Loyalty level with Total revenue and Payment method

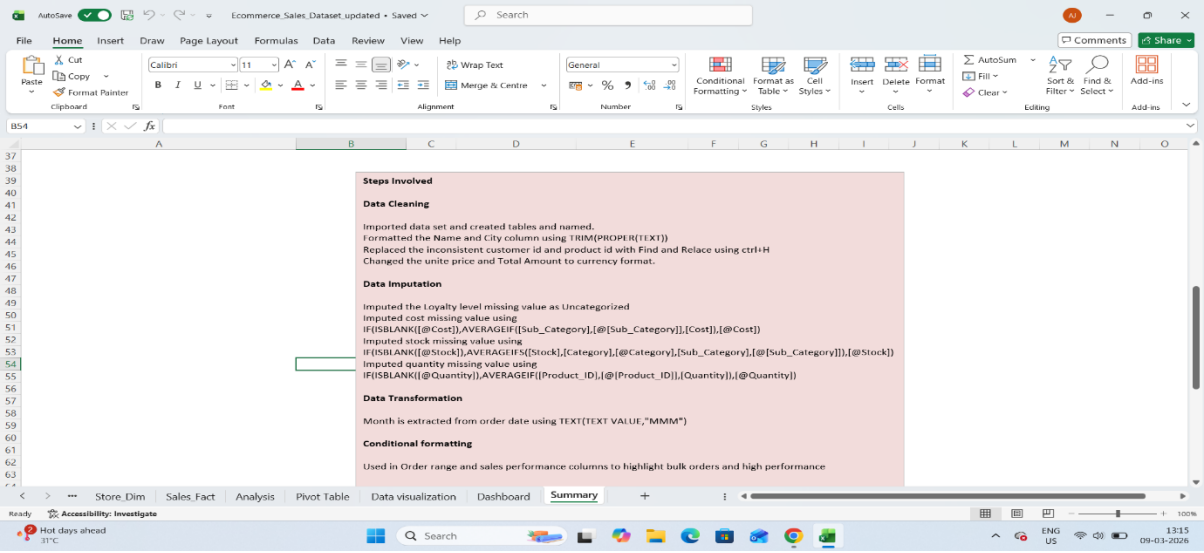
The Platinum customers are having the highest transaction values and most commonly Paypal method of transaction is used widely among customers.

4.Monthly sales distribution

Looking at the month sales and products

The March month has the highest contribution in total revenue of $210704.52

Highly purchased product is Men's casual sneakers



Steps Involved

Data Cleaning

Imported data set and created tables and named.
Formatted the Name and City column using TRIM(PROPER(TEXT))
Replaced the inconsistent customer id and product id with Find and Relace using ctrl+H
Changed the unite price and Total Amount to currency format.

Data Imputation

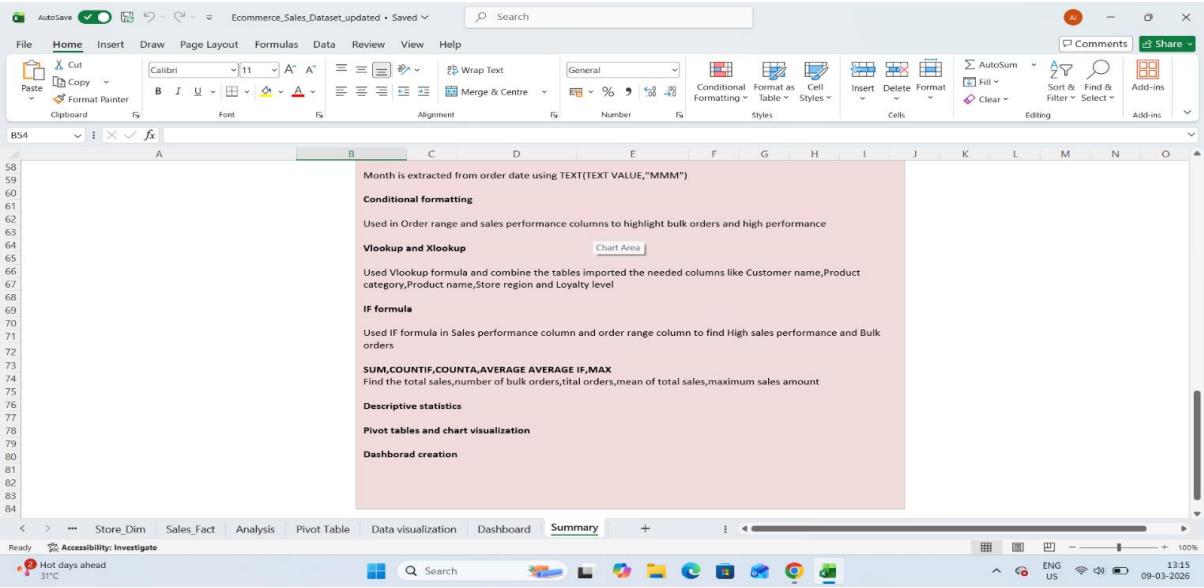
Imputed the Loyalty level missing value as Uncategorized
Imputed cost missing value using
IF(ISBLANK([@Cost]),AVERAGEIF([Sub_Category],[@[Sub_Category]],[Cost]),[@Cost])
Imputed stock missing value using
IF(ISBLANK([@Stock]),AVERAGEIFS([Stock],[Category],[@Category],[Sub_Category],[@[Sub_Category]]),[@Stock])
Imputed quantity missing value using
IF(ISBLANK([@Quantity]),AVERAGEIF([Product_ID],[@[Product_ID]],[Quantity]),[@Quantity])

Data Transformation

Month is extracted from order date using TEXT(TEXT VALUE,"MMM")

Conditional formatting

Used in Order range and sales performance columns to highlight bulk orders and high performance



Month is extracted from order date using TEXT(TEXT VALUE,"MMM")

Conditional formatting

Used in Order range and sales performance columns to highlight bulk orders and high performance

Vlookup and Xlookup

Chart Area

Used Vlookup formula and combine the tables imported the needed columns like Customer name,Product category,Product name,Store region and Loyalty level

IF formula

Used IF formula in Sales performance column and order range column to find High sales performance and Bulk orders

SUM,COUNTIF,COUNTA,AVERAGE AVERAGE IF,MAX
Find the total sales,number of bulk orders,tital orders,mean of total sales,maximum sales amount

Descriptive statistics

Pivot tables and chart visualization

Dashborad creation